

CLOUD COMPUTING PROJECT

AWS S3 Cloud Storage System

Scalable, Secure & Interactive Object Management

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Project Overview & Objectives

What is AWS S3?

- **Amazon Simple Storage Service:** Scalable object storage in the cloud.
- **99.999999999% Durability:** Built for mission-critical data protection.
- **Global Access:** Direct REST API access for fast file transfer.

Project Scope

- **Full File Lifecycle:** Upload, search, view, and delete operations.
- **User Management:** Secure authentication and personal vaults.
- **AWS Integration:** Real-time sync with AWS S3 cloud buckets.

DATA STORAGE ARCHITECTURE FLOW

1. User Request
Client uploads file



2. AWS API Gateway
Auth & Routing



3. S3 Bucket Storage
Object key mapping



4. Global Availability
Replicated across AZs

System Architecture & Tech Stack

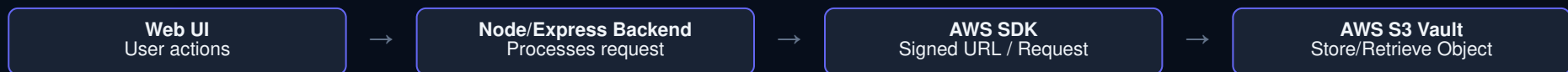
Frontend & Auth Layer

- **User Dashboard:** Interactive UI for registration & login.
- **Session Security:** Encrypted token auth for user sessions.
- **File Control:** Live view and search interface for files.

Backend & Cloud SDK

- **AWS SDK Core:** API bindings to communicate with S3.
- **Object Management:** Unique key generation for stored assets.
- **Validation Engine:** Verifies payload limits and access rights.

SYSTEM INTERACTION FLOWCHART



Security & Access Control

Identity & IAM

Strict IAM role-based policies restrict bucket access to validated credentials only.

Data Encryption

Server-Side Encryption (SSE-S3) protects files automatically at rest using AES-256.

SSL/TLS In-Transit

HTTPS endpoints enforce end-to-end encryption during all network file transfers.

SECURITY ENFORCEMENT PIPELINE

Incoming File
TLS encrypted



IAM Check
Verify bucket policy



SSE-S3 Engine
AES-256 encryption



Protected Object
Secure in S3

Project Execution Workflow

1. User Onboarding

Register account & authenticate login credentials to receive session access.

2. Vault Dashboard

Welcome page displays personal cloud storage statistics & active files.

3. Cloud Operations

Upload new files, search assets, modify details, or remove S3 objects.

APPLICATION EXECUTION LIFECYCLE

Register / Login
Auth portal



Welcome UI
User dashboard



Upload / Search
File interaction



Modify / Delete
S3 sync complete

SLIDE 06

Technical Highlights & Benefits

Pay-As-You-Go

Zero upfront cost; pay exclusively for active cloud storage consumption.

Fast Metadata Search

Indexed file parameters enable instant retrieval of files across large buckets.

Infinite Elasticity

Scales seamlessly from megabytes to petabytes without hardware limits.

CLOUD VS TRADITIONAL STORAGE VALUE FLOW

Local Hardware
Fixed capacity & risky



AWS S3 Migration
Cloud integration



Automated Scale
Infinite storage space



Cost Savings
Pay only for usage

Thank You

Thank you for watching and listening patiently.
Your time and consideration throughout this presentation are greatly
appreciated.